CISC 211 1-3/5

Encrypting the message

1. message: circle

2.

c	i	r	c	l	e
99	105	114	99	108	101
64+32+2+1	64+32+8+1	64+32+16+2	64+32+2+1	64+32+8+4	64+32+4+1
0110 0011	0110 1001	0111 0010	0110 0011	0110 1100	0110 0101

3. key: square

s	q	u	a	r	e
115	113	117	97	114	101
64+32+16+2+1	64+32+16+1	64+32+16+4+1	64+32+1	64+32+16+2	64+32+4+1
0111 0011	0111 0001	0111 0101	0110 0001	0111 0010	0110 0101

4. XOR
Binary:

	0110	0011	0110	1001	0111	0010	0110	0011	0110	1100	0110	0101
XOR	0111	0011	0111	0001	0111	0101	0110	0001	0111	0010	0110	0101
=	0001	0000	0001	1000	0000	0111	0000	0010	0001	1110	0000	0000
	1	0,	1	8,	0	4+2+1,	0	2,	1	8+4+2,		

5. Hexadecimal: 10 18 07 02 1E 00

Decrypting the message

1. 0001 0000 0001 1000 0000 0111 0000 0010 0001 1110 0000 0000
XOR 0111 0011 0111 0001 0111 0101 0110 0001 0111 0010 0110 0101

: 0110 0011 / 0110 1001 / 0111 0010 / 0110 / 0011 / 0110 1100 / 0110 0101
4+2 2+1 4+2 8+1 4+2+1 2 4+2 2+1 4+2 8+4 4+2 4+1

Hexadecimal: 63 69 72 63 6C 65 ASCII: 99 105 114 99 108 101

2. decryption matches the original ASCII (circle)

3. If the key is shorter than the message:
   - loop the key over & over until it reaches the message character count

   If the key is longer than the message:
   - cut it off at the number of message characters